

Pair Table Method (easy to program)

The reduction consists of two algorithms:

- One that builds a pair table for the non-reduced FA.
- The other that marks certain rows.

At the end of the second algorithm, each row that is not marked is associated with a pair of equivalent states; each row that is marked is associated with a pair of distinguishable states.

A **feasible state-pair** is a pair of states that conceivably could be equivalent upon a cursory examination of the FA's transition table. A pair of states (p, q) is a member of the feasible state-pair set if:

1. $\{p, q\}$ is a subset of either F or $Q - F$, and
2. for every input symbol "a", $\delta(p, a)$ is $\emptyset$ iff $\delta(q, a)$ is $\emptyset$.

Construction of the Pairs Table

Given a pair (p, q) associated with a row, then the table entry for symbol "a" is a pair (p', q') where $p' = \delta(p, a)$ and $q' = \delta(q, a)$. That is, simply list the pair of states to which (p, q) transfers on each input symbol. A pair is unordered, e.g. the pair (i, j) is equivalent to (j, i) .

Marking Distinguishable State-pairs

A feasible state-pair is marked if there exists a transition to a state-pair (p, q) such that:

1. p and q are different, and
2. (p, q) is either marked or **not** among the feasible state-pairs.
3. repeat until no more state-pair can be marked.